

The instrument is not the bottleneck: an evidence-first observatory for Belgrade, and what it refuses to say

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Prepared for *Creating sustainable commUNiTy*, 6th International Conference on Sustainable Environment and Technologies, 25 September 2026. This is a scientific paper by its authors. The work runs on the authors' own equipment, is non-commercial, and does not speak for the university or bind it. **Svemir** — the authors' local AI infrastructure — is recorded as a verified contributor, not as an author.

Every figure in this paper is printed by `tools/paper_numbers.py`, which reads the record rather than a summary of it. The figures below were produced on 9 September 2026 at 16:34 UTC. The system, its data, its refusals and this paper's numbers are public.

Abstract

Cities are told they need more sensors. Belgrade already has a state air-quality network, meteorological and hydrological stations, airport observations, utility outage tables, parking counts and citizen sensors, and a resident still cannot establish what the air is doing nearby, how certain anyone is, and when it was measured. We argue the binding constraint is not sensing but the defensible record, and we test the argument by building one. A standard-library collector reads 28 permitted public sources every 10 to 60 minutes; every value carries its source, three times, a unit, one of five evidential states and the captured permission that allowed the read; absence is recorded with its reason; corrections are appended and never erased. Of 211 registered access routes, 14 publishers refused and are honoured by name. Mapped onto ISO 37120, exactly one of nineteen themes is reachable by real-time measurement. Small local models narrate the record behind a validator, which we evaluate adversarially rather than demonstrate: it admitted 17 of 36 unsupportable utterances before hardening and 3 after, and we name the three it still cannot catch.

Keywords: urban observatory; data provenance; evidential state; open data; language-model grounding; ISO 37120

1 • The gap

A resident of Belgrade who wants to know whether the air near their street is bad this afternoon can find numbers within a minute. What they cannot find, on any public surface, is the answer to four questions that decide whether those numbers mean anything: where did this come from, when was it measured rather than when was it

displayed, how certain is it, and what is missing. The last question is the one that is never answered, because absence has no pixel.

The reflex response to this situation is to propose more instruments. We think the reflex is wrong, and the evidence for that comes from the deployments that already did it. The Newcastle Urban Observatory — on its own account more than 3,600 sensor streams across 65 variables, on the order of 10,000 observations per minute, over ten billion observations collected — reports that its own bottleneck lies elsewhere: analysis remains manual, time-consuming and expensive, and few people actually want data (James et al., 2022). The same paper records what dashboards rarely do: a sensor developing a fault and reporting excessive values, air sensors classified as indicative and permitted to sit within 50 % of a reference, maintenance becoming a substantial burden past a certain fleet size. Scale did not produce insight; it produced a maintenance problem and a haystack.

The critical literature names the missing layer precisely. Among the six key issues Kitchen and McArdle identify for city dashboards, *veracity and validity* is the one where data quality metadata is rarely provided while the dashboard demands trust without transparency; under *scope and access*, their own Dublin audit found only 10 of 37 sustainability indicators available at an appropriate scale (Kitchen & McArdle, 2016; Kitchen, Lauriault & McArdle, 2015). What is missing from the public picture of a city is not sensing. It is the record about the sensing.

The new element is that the same gap now has a machine standing in it. Huang (2025) defines *geo-hallucination* as model output that is spatially incoherent or fabricated while appearing authoritative, arising from the model's reasoning rather than from measurement, and treats it as an epistemic and socio-political risk for planning and governance; the remedy proposed is grounding, critical literacy and a renewed commitment to evidence with human oversight. A language model placed over a city's data does not fail by inventing a sensor. It fails by writing a fluent sentence about one that is silent.

Our claim, stated so it can be refuted. A city becomes knowable not by acquiring instruments but by keeping a record in which every value can be traced, dated, doubted and, where necessary, contradicted. We do not claim that better bookkeeping improves any decision — we have run no user study and make no such claim. We claim the weaker and more testable thing: a decision cannot be well made on a number whose age, origin and permission are unknown, and today those are unknown for almost everything a Belgrade resident can see.

2 • Method: the record before the reading

Permission is bytes, not a belief. Before the first read of any source we store its rules — robots.txt, terms of use, licence page — byte for byte, with the fetch time and a sha256, and re-check weekly. A source enters the collector's list only if that capture exists; the collector re-checks the gate on every tick and refuses otherwise. Fourteen publishers said no. They are recorded by name with the clause that says so and are never read, including one whose material would fall outside copyright in any case as an official document. Serbia has **no text-and-data-mining exception**, so no part of this design

relies on one: we take a headline, a link and a publication time, never an article body, never an image.

Three times, not one. Every row carries the *phenomenon time* (when the world was as the value says), the *result time* (when the source says it computed or published the value), and the *reception time* (when the bytes reached us). Only the third is ours to know for certain. Conflating them is the default failure of live displays: a value that arrived four minutes ago is shown as a value that happened four minutes ago.

Five evidential states. `observed` (measured, with its own measurement time), `untimed` (a real value whose measurement instant the source does not publish), `estimated`, `forecast`, `unavailable`. The `untimed` state is, to our knowledge, not standard practice on public city displays, and it is not a technicality. Belgrade’s parking source publishes counts with no measurement time at all. The honest rendering is therefore not a number but a number with a smeared age, and our drawings smear it.

Absence is a positive record. A source that says nothing produces a row that says nothing was received, with the reason. Missing is never zero. This single rule is what most distinguishes the record from a dashboard, and it costs storage rather than saving it.

Corrections are appended. Fourteen corrections (C-001 ... C-014) are recorded; none edits the text of the observation it corrects. The newest ledger line for a source wins; the older lines stay.

The whole collector is Python standard library: no dependency, no external service, one machine. That is a research choice, not asceticism — a record whose provenance depends on a supply chain we cannot audit is not a provenance record.

3 • What Belgrade actually publishes

Quantity	Value (9 September 2026, 16:34 UTC)
Registered access routes	211 , each with a status; 14 refused and honoured, 3 awaiting a decision
Permission evidence	188 sources with stored evidence · 313 captures · 169 access checks passed, 11 refused, 4 awaiting, 4 incomplete, 26 undocumented
Sources actually polled	28 of 31 listed (3 disabled because their publisher refused); cadence 10–60 min; 20 of them news
Rows on disk	49,379 across 28 source files
Hourly history	23 h , 843 series — 794 on a measured clock, 49 on reception only
Corrections	14 , appended, never edited
Offline tests	155 , all passing

Two of these numbers deserve emphasis. **Twenty-six sources are undocumented** — we read them, and we cannot yet show the paperwork to the standard the other 169 meet. It

is in the table because leaving it out would make the table a claim rather than an audit. And **49 of 843 series have no measured clock**: for one series in seventeen we know the value and not its age, and the record says so on every surface where it appears.

Against a standard, not against ourselves. An audit of openness performed with one’s own categories measures one’s own appetite. We therefore mapped the record onto ISO 37120:2018, the indicator set a city would actually be held to (19 themes). The result is the most useful negative finding in this paper:

- **1 theme of 19** — Environment and climate change — can be reached at indicator level by real-time measurement of public sources, and only after a year of record.
- **6 themes** are touched by something real that the standard does not ask for: a river level, a planned outage, a parking count, a service notice.
- **10 themes** are annual administrative statistics — unemployment, enrolment, recycling share, wastewater treated — that no instrument reports and no cadence will produce.
- **2 themes** we refuse by design: Health, and Safety, where the interior ministry is among the fourteen refusals and where person-level data would change the legal basis of the whole record.

The gap between what a city can measure continuously and what a city is asked to report is not a gap in instruments. Sensors do not close it. It is a gap in what is written down, by whom, and how often — which is this paper’s argument, arrived at from the standard’s side rather than from ours.

4 • The gate, measured rather than demonstrated

Three small language models run locally over the record and comment on it in Serbian. They never touch a source; they read a digest in which every fact is numbered and every number the machine may ever utter is present. Between them and the public page stands a validator: it admits only numbers present in the digest, requires a citation inside the sentence, refuses a forecast stated as fact, checks the language and the script, and refuses a restatement of the conversation. Every utterance is machine-marked as AI-generated (EU AI Act, Art. 50) and is never presented as a measurement.

Until this week we reported that gate by its refusal count — 102 refusals across 222 utterances, 45.9 %. **A refusal count says how often a gate fired. It says nothing about whether it fired at the right things.** So we built a fixed adversarial set: 44 hand-labelled utterances over one synthetic digest, 36 labelled *reject* (a careful reader could not verify them from the digest) and 8 labelled *accept*. No model is called; the only thing under test is the code that decides.

The first result was bad, and is the reason this section exists.

	organ v0.3.5	organ v0.4.0
False accepts (of 36)	17 — rate 0.472	3 — rate 0.083
False rejects (of 8)	1	1

The arithmetic checks — is this number in the digest, is there a citation, is this a naked prediction — caught every attack made of tokens and **missed almost every attack made of meaning**. Among the seventeen: a value with no measurement time presented as the present state; silence read as nothing having happened; a similarity turned into a confirmed event; a cause the record does not carry; a superlative over a period outside the window; guidance to residents; a unit off by a factor of a thousand; words put into a state agency's mouth. Each of these would have been published, in Serbian, under a label saying the record supports it.

We then wrote the guards that a rule can honestly express — a number must come from a fact the sentence actually cites rather than from anywhere in the digest; a fact flagged as having no measurement time may not be spoken of in the present tense; a silence fact may not be followed by an assertion of nothing; a cited fact's units bound the units the sentence may use; a rendering may not introduce a number the sentence it renders does not contain — and re-ran the same set. Fourteen of the seventeen close. The remaining three are named rather than hidden:

1. **number rebinding** — a number taken from the correct cited fact and attached to the wrong quantity inside it (8 instruments becomes 8 $\mu\text{g}/\text{m}^3$);
2. **measurement-time drift** — a value with a real measurement time an hour old, called current;
3. **voice meaning drift** — a Serbian rendering with the same numbers, the same citation and correct ekavica, saying the opposite of the English it renders.

All three need a reader, not a rule. They are pinned by a regression test, so a future edit that reopens one of the fourteen fails the suite, and a future edit that closes one of the three fails it too — the correct response then being to move that family out of the residue and out of this paragraph, in that order.

What this evaluation is not. One annotator wrote both the validator and the set. The attacks are the ones its author could imagine in one sitting; there is no inter-rater agreement, no baseline system, and no second observatory to compare against. A false-accept rate measured this way is a **lower bound**. We report it because a lower bound measured is worth more than a refusal count recited, and because the honest form of the contribution is now visible: *the contribution is the gate and the record, not the model* — and the gate, measured, was 47 % permeable before we looked.

5 • The failure that reported success

On 8–9 September the task that publishes this system's own record ran roughly a hundred times over seventeen hours. Its status was Ready, its last result was 0, and a log line appeared on every run. It published nothing. A batch file that calls another batch file without `call` never returns, so lines two and three of the publish script had not executed since the file was last edited. The proof is in the log: zero occurrences of the string that the publish step writes, before the fix; one immediately after.

The failure was invisible precisely because the self-report was healthy. The system was doing what it accuses dashboards of doing: trusting a receipt instead of the artefact. The fix in the scripts is trivial; the correction that matters is the check now added to the

verifier — the published artefact’s age must be under 25 minutes, and the exit code is not evidence of anything.

An exit code of 0 is evidence that something finished. It is never evidence that it happened. A paper about evidence should say when it caught itself.

6 • Lawfulness, and where it is thin

The record is built on captured permission, short quotation (Copyright Act, Art. 49), and official materials outside copyright altogether (Art. 6(2)); no TDM exception is relied on, because Serbia has none. Nothing is taken from behind a login, a paywall or a prohibition, and no technical protection is circumvented anywhere in the system.

The news layer processes personal data — a headline names people — and this is the part that was thin. As of the date of this paper it has: a written legal basis (scientific research, ZZPL Art. 12 with the safeguards of Art. 92), a written record of processing activities (Art. 47), a public notice relying on the disproportionate-effort exception for data not collected from the subject (Art. 24), a standing objection address on every public page (Art. 37) acted on within 30 days, and a **retention rule**: headline text is erased after 90 days, the row surviving with its source, link and times as evidence that something was published at that minute at that address. Enforcement is code, not intention: a tool computes what is due, a ledger records every erasure with the file’s hash before and after, and the first erasure falls due on **7 December 2026**.

Two weaknesses remain and are stated rather than managed. The published snapshot is served from outside Serbia, which is a transfer question we answer as publication of already-published material within a scientific work, and which a regulator might answer differently. And the sources have not been notified individually: their published rules are honoured, which is lawful, but weaker than being told. Letters to the fourteen refusing publishers and to the agencies are drafted and await signature. Until they are sent, our position is “they did not forbid it”, not “they agreed”.

7 • Limitations

- **Twenty-three hours of history.** Any statement about how the city’s values move over time is, today, unsupported by our own record. The site reports the record’s age everywhere it appears, and this paper claims nothing that a day cannot carry.
- **The gate evaluation is a lower bound** (§4), on one machine, with three small models, by one annotator.
- **The gate changed on the day this paper was written.** Refusal counts from before 9 September and after it are not comparable, and are not compared.
- **No user study.** Nothing here shows that a better record produces a better decision. The claim is the negative one, and it stays negative.
- **Twenty-six undocumented sources.** The audit is honest about its own hole; the hole is still a hole.

- **The ISO mapping is thematic.** The standard's text is not open. We map to its 19 published themes and compute no ISO indicator, and we claim no conformity.

8 • Conclusion

We set out to test whether a city becomes knowable through instruments or through the record kept about them, and we did it by building the record for one city in public.

Twenty-eight permitted sources produce 49,379 rows in which every value carries its origin, three times, a state and its permission; fourteen publishers' refusals are part of the data; absence is written down; corrections are appended; and one machine layer that could have made all of it fluent and false was measured instead of demonstrated, at 47 % permeable, then hardened to 8 % with three named failures it cannot reach.

Against ISO 37120, the whole apparatus reaches one theme of nineteen. We take that as the paper's strongest result rather than its weakest: the distance between what a city can sense and what a city is asked to know is not crossed by sensing. It is crossed by writing things down in a way that can be checked — including the parts that are missing, refused, or wrong.

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Honest verdict

- The paper's spine is now a **measurement, not a description**: 0.472 → 0.083 on a fixed set, with three named residues. Before today the same section was an anecdote with a refusal count in it.
- The ISO result (1 of 19) is the finding a reviewer will remember, and it is the one we did not set out to produce.
- The weakest sentence in this paper is still “the contribution is the gate”, and §4 now concedes exactly why: one annotator, no baseline, a lower bound. A reviewer who presses there is right.
- Twenty-three hours of history remains thin, and no amount of writing improves it — only time does.
- Two things this paper cannot fix by working harder: the letters need a signature, and the late-submission question needs to be asked. Both are the authors', not the system's.